FAST FOCUS – VERSIONING ASP.NET CORE SERVICES

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https://github.com/skimedic/presentations

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Phil.About()

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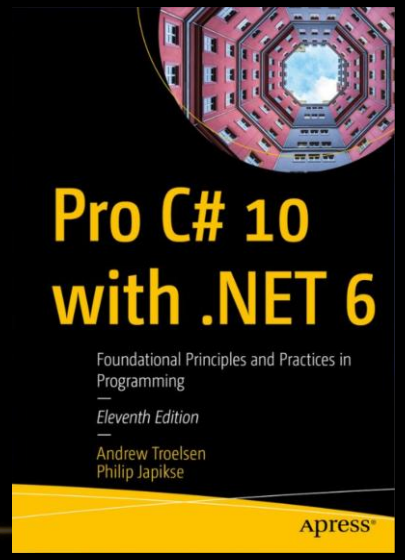

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VERSIONING OPTIONS

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VERSION FORMATS

- Services are versioned using the Major.Minor versioning scheme
  - 1.0, 2.0
- Services can opt for only the Major version - the ".0" is implied
  - v1 => v1.0
- Status (RC, Alpha, Beta, etc.) can be specified after the Minor version
  - 1.0-Alpha
- Grouping using YYYY-MM-DD
  - 2018-06-12.1.0-RC

https://github.com/Microsoft/api-guidelines/blob/vNext/Guidelines.md

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VERSIONING OPTIONS

- Embed the version after the service root
  - https://www.skimedic.com/api/v1.0/classes
- Use a query string parameter
  - https://www.skimedic.com/api/classes?api-version=1.0
- HTTP Headers (Not compliant with MS REST Guidelines)
  - api-version:2.0
- Use Media Type (Not compliant, but generally accepted)
  - Content-Type: application/json;v=2.0 (or Accept)
    - Content-Type wins if both are present

https://github.com/Microsoft/api-guidelines/blob/vNext/Guidelines.md

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GROUP VERSIONING

- Group Versioning is an optional feature
  - Defined using the YYYY-MM-DD format
  - Does not replace the Major.Minor version format
- Allows for logical grouping of API
  - Developers can lookup a single version and use it across related end points
- Can cause confusion due to reusing versions

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VERSIONING ASP.NET CORE WEB SERVICES

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OVERVIEW

- Add Package Asp.Versioning.Mvc
- In Program.cs
  - Add call to builder.Services.AddApiVersioning().AddMvc()
  - Add call to builder.Services.AddProblemDetails
- Use Attribute Routing for controllers
  - If using URL versioning, update the route
- Add version attributes
- Getting the version requested:
  - HttpContext.GetRequestedApiVersion || Model Binding

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ADDING VERSIONING TO CONTROLLERS

- Use the ApiVersion attribute to add versioning

```
//Query String and Media Type
[ApiVersion("2.0")]
[Route("api/helloworld")]
public class HelloWorld2Controller : Controller
{
    …
}
//URL Versioning
[ApiVersion(1.0)]
[Route("api/v{version:apiVersion}/[controller]")]
public class HelloWorldController : Controller
{
    …
}
```

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VERSION INTERLEAVING

- Use the ApiVersion/MapToApiVersion attributes to add versioning

```
[ApiVersion("2.0")]
[ApiVersion("3.0")]
[Route("api/v{version:apiVersion}/helloworld")]
public class HelloWorld2Controller : Controller
{
    [HttpGet]
    public IActionResult Get() => "Hello world v2!";

    [HttpGet, MapToApiVersion("3.0")]
    public IActionResult GetV3() => "Hello world v3!";
}
```

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DEPRECATING VERSIONS

- Add Deprecated to the ApiVersion attribute

```
[ApiVersion("1.0", Deprecated = true)]
[Route( "api/[controller]" )]
public class HelloWorldController : Controller
{
   //omitted
}
```

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ADVERTISE VERSIONS

- Advertise versions outside of the controller heirarchy

```
[AdvertiseApiVersions("2.0","3.0")]
[AdvertiseApiVersions("1.5",Deprecated=true)]
[ApiVersion(1.0)]
[Route( "api/[controller]" )]
public class HelloWorldController : Controller
{
   //omitted
}
```

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VERSION NEUTRALITY

- Use the ApiVersionNeutral attribute to expose an endpoint to all versions

```
[ApiVersionNeutral]
[Route("api/v{version:apiVersion}/[controller]/[action]")]
public class HealthController : Controller
{
    [HttpGet]
    public IActionResult Ping(ApiVersion versionFromDI)
    {
       ApiVersion version = HttpContext.GetRequestedApiVersion();
    }
}
```

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GETTING THE REQUESTED VERSION INFORMATION

- Use the GetRequestedApiVersion or ModelBinding to return the requested version information

```
[ApiVersion("1.0")]
[ApiVersion("2.0")]
[Route("api/v{version:apiVersion}/[controller]/[action]")]
public class DifferentVersionsController : Controller
{
   [HttpGet]
   public string RequestedApiVersion() =>
      JsonConvert.SerializeObject(HttpContext.GetRequestedApiVersion());

   [HttpGet] //3.0+
   public string Get(ApiVersion apiVersion)
      => $"Controller = {GetType().Name}\nVersion = {apiVersion}";
}
```

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Questions?

Thank You!

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